

REPORT METADATA AND SURVEY DETAILS

Report reference: MRPL-DEMO-2026-0007 (synthetic sample data).

Location: Refinery Unit 3, crude distillation area.

Date of survey: 12 February 2026. Surveyor: A. Sharma, Level II.

Document control: revision 1, distribution restricted to unit records.

SECTION 1 — EQUIPMENT CONDITION

Safety-critical finding requiring attention: pump P-204 casing shows moderate surface wear and coating breakdown, and the flange gasket seating is aged.

No active leakage was observed at the time of survey; the condition should be monitored to prevent deterioration.

Recommendation: monitor the coating and re-inspect at the next scheduled maintenance window.

SECTION 2 — PIPELINE & JOINT INSPECTION

Safety-critical finding requiring attention: weld seam WJ-42 on line 6-P-118 exhibits surface irregularity and a reduced wall thickness measured by ultrasonic testing.

Continued thinning of the joint could compromise pressure containment under operating conditions and is safety-relevant.

Recommendation: schedule a detailed fitness-for-service assessment of the affected joint.

SECTION 3 — STRUCTURAL & EQUIPMENT INTEGRITY

Safety-critical finding requiring attention: significant corrosion and section loss were observed on the support structure and the lower shell of vessel V-311 near the flange connection.

The affected support is load-bearing; prompt engineering evaluation of the corrosion is required to assure integrity.

Recommendation: schedule a detailed inspection and maintenance assessment, and restrict load until the corrosion is evaluated.

SECTION 4 — SAFETY OBSERVATIONS

Safety-critical finding requiring attention: the emergency access walkway has a corroded handrail and a loose grating panel, presenting a fall hazard to personnel.

A fire-water isolation valve was found partially obstructed, which could delay emergency response.

Recommendation: rectify the walkway hazards and clear the valve obstruction without delay.